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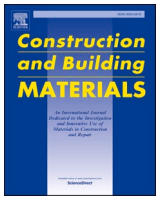

A time-series deep learning model for predicting concrete shrinkage and creep verified with in-situ and laboratory test data

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ABSTRACT

This paper proposed a time series prediction method for concrete shrinkage and creep based on deep learning and the accuracy of the method was verified by field tests. Based on the experiment data in the Northwestern University (NU database), a deep learning model considering the material, environment, and history was constructed by integrating Gated Recurrent Units and Self-Attention mechanism (GSA model). To validated the model, short-term indoor and outdoor field tests of concrete shrinkage and creep were conducted. The proposed GSA model outperformed the LSTM model for both creep and shrinkage predictions. Moreover, the comparison between the GSA model prediction and the outdoor test data indicated that the model can accurately capture the real-time outdoor temperature and humidity variation through historical data. To predict the ten-year shrinkage and creep of concrete, a recursive multi-step method was proposed to extrapolate the short-term predictions of the GSA model. The effectiveness of the recursive multi-step method was validated against the long-term test data in the NU database. Overall, the proposed GSA model demonstrated high accuracy in both short-term and long-term predictions.

1. Introduction

Concrete creep refers to the deformation of concrete over time under constant load, while concrete shrinkage is the volume reduction of concrete due to cement hydration reaction or environmental factors such as drying without external force. Shrinkage and creep are inherent characteristics of concrete, which, if not properly controlled, can lead to a series of structural failures. These include long-term deflection and web cracking in prestressed continuous rigid-frame bridges [1], long-term failure of concrete containment structures in nuclear power plants [2], and cracking of concrete components [3]. Consequently, the phenomena of shrinkage and creep present a challenging issue in the field of civil engineering, warranting significant research attention.

Accurate prediction models are crucial for addressing the issues of shrinkage and creep. To date, many explicit regression models have been developed and successfully applied in engineering practice. Among them, the fib2010 model proposed by the European Concrete Committee and the International Federation for Structural Concrete (CEB-FIP) [4] has a wide range of temperature applicability [5]. The B4 model by Bazant and his colleagues [6], which considers numerous calculation parameters and steps, offers higher accuracy and broader applicability, allowing for the prediction of concrete shrinkage and creep beyond typical ranges [7,8]. The AASHTO model developed by the American Association of State Highway and Transportation Officials [9] is specifically designed for high-strength concrete, making it more suitable for predicting the shrinkage and creep of such materials [10]. However, due to the highly complex and nonlinear nature of concrete shrinkage and creep, coupled with their significant temporal randomness [11], these models exhibit certain limitations in predicting long-term deformation.

With the advent of artificial intelligence, machine learning has emerged as an effective tool for addressing the nonlinear behaviors of concrete [12–16]. The progressive enhancement of experiment data in the Northwestern University (NU database) has provided a robust foundation for AI-based prediction of concrete shrinkage and creep [17].

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Taha et al [12]. developed an artificial neural network (ANN) with a hidden layer of six neurons to predict masonry creep. However, this study considered only four parameters—stress level, relative humidity, loading time, and creep measurement time—and utilized merely 14 data sets for model validation, resulting in limited generalizability. El-Shafie et al [15]. introduced a creep prediction model based on nonlinear autoregressive exogenous input (Narx), which showed improved accuracy over ANN. Nonetheless, these studies suffer from small data sets and limited model applicability. [18,19] included more factors and leveraged larger data sets but focused solely on basic creep prediction, neglecting drying creep and shrinkage strain.

Due to the powerful feature extraction capabilities, deep learning techniques have been widely applied in fields such as image recognition, natural language processing, and time series forecasting [20]. The time-dependent behavior of concrete shrinkage and creep represents a typical time series, making it suitable to apply deep learning (DL) models for time series to accurately capture the relationships between material characteristics, environmental factors, and historical data with shrinkage and creep.

Chen [21] considered the influence of material characteristics and environmental parameters, utilizing an LSTM model to train on the NU database, and found that the LSTM model's prediction outperformed the B3 model. Wang [22] used experimental data to develop a concrete creep model based on an LSTM network, validating the effectiveness of the method in capturing the time-dependent changes in concrete creep through single-step predictions compared with experimental values. However, this study neglected the impact of material characteristics and environmental factors on creep and did not address long-term creep prediction methods. Zhu et al [23]. proposed a shrinkage and creep model based on convolutional neural networks (CNNs), validated using 906 sets of creep experimental data and 1114 sets of shrinkage experimental data from the NU database. The validation results indicated that the CNN model's accuracy far surpassed that of the B4 model. However, the input parameters considered only material characteristics and environmental factors, neglecting the influence of historical shrinkage and creep data on current predictions.

In practical engineering, most concrete structures are constructed in natural environments. Research has shown that variations in temperature and humidity in natural environments significantly affect shrinkage and creep. Ahari et al [24]. studied the variation of creep strain under different humidity through experiments, and the results showed that when the relative humidity increased from 50 % to 80 %, the creep coefficient decreased by 30 %. Parrott [25] and Barr et al [26]. conducted experimental studies on shrinkage and creep under varying temperatures and relative humidity, and found that the shrinkage and creep strains in natural environments and controlled environments were very different.

Since the traditional explicit prediction models are based on experiments conducted in indoor, constant temperature and humidity conditions, these models typically require enhanced underlying formulas to improve prediction accuracy when dealing with shrinkage and creep in natural environments. Fahmi et al [27]. conducted concrete shrinkage and creep tests under varying temperature and relative humidity conditions and proposed a shrinkage and creep calculation method addressing temperature changes based on the principle of superposition [28]. Zhan et al [29]. developed a correction model for shrinkage and creep in natural environments based on the superposition principle and the CEB90 code, and validated the model through experiments. Bazant et al [30]. proposed the Microprestress-Solidification (MPS) theory based on consolidation theory to study drying creep. The MPS theory takes into account the effects of changing temperature and humidity by modifying the age of concrete in calculations.

Currently, predicting shrinkage and creep under natural environmental conditions remains highly challenging. To further explore the application of deep learning in predicting shrinkage and creep in such environments, this paper considers material characteristics, environmental parameters, and historical values to propose a time series deep learning-based prediction model. This model is validated through experiments conducted under natural conditions. The remainder of the paper arranged as follows: In Section 2, based on the NU database, a Gated Recurrent Units (GRU) and self-attention mechanism deep network model (GSA model) is constructed and its performance is compared with the LSTM model. Section 3 presents the results of field shrinkage and creep tests. Section 4 conducts single-step predictions of shrinkage and creep using the proposed model and verifies its accuracy by comparing it with standard models. Section 5 extrapolates the ten-year shrinkage and creep development trend based on short-term experimental data using a recursive multi-step method.

Table 1
Input parameters for the Model.

No.	Symbol	Description	Unit	Creep Database Range	Shrinkage Database Range
x1	t	Age	lg (day)	[0, 3.56]	[0, 3.56]
x2	J ε	Historical creep compliance/ shrinkage strain	10^{-6}/ MPa 10^{-6}	[0427]	[−2058,152]
x3	type	Cement type	-	[1,2,3,4]	[1,2,3,4]
x4	c	Cement content	kg/ m3	[214,1314]	[222,1616]
x5	w/c	Water-cement ratio	-	[0.2, 0.9]	[0.24, 0.85]
x6	a/c	Aggregate-cement ratio	-	[0.5, 9]	[0, 8.32]
x7	V/S	Volume-surface ratio	mm	[5.3, 126.3]	[15.56, 135.36]
x8	f28	28-day strength	MPa	[10.8, 54.88]	[10.8, 129]
x9	E28	28-day elastic modulus	MPa	[14535,50800]	[14535,64500]
x10	RH	Relative humidity	%	[3100]	[10,100]
x11	T	Temperature	℃	[18,130]	[17,130]
x12	t0 ts	Loading age/ drying age	day	[0.66, 3295]	[0.42, 625]

2. Deep-learning-based time-series prediction model

2.1. Model parameter

Before establishing the prediction models for concrete creep and shrinkage, it is necessary to determine the input parameters of the model. Since the B4 model is developed based on the widely accepted micro prestress-solidification theory, it is reasonable to select the model's input parameters on this basis [21]. conducted a principal component analysis of the shrinkage and creep data from the NU database. The results indicate that the most influential factors on shrinkage and creep are the material characteristics of the concrete, including strength, elastic modulus, and mix proportions. The geometric characteristics of the components and environmental temperature and humidity factors are secondary.

Based on the parameter selection of the B4 model and the analysis results from [21], twelve feature parameters were selected as input parameters for the model, as shown in Table 1. Among these, x2 represents the historical data of shrinkage and creep. Since concrete shrinkage and creep form a time series, the patterns in the historical data can provide clues about potential future occurrences.

2.2. Model architecture

A deep neural network model combining GRU and Self-Attention was implemented with Python. The GRU is a type of gated recurrent neural

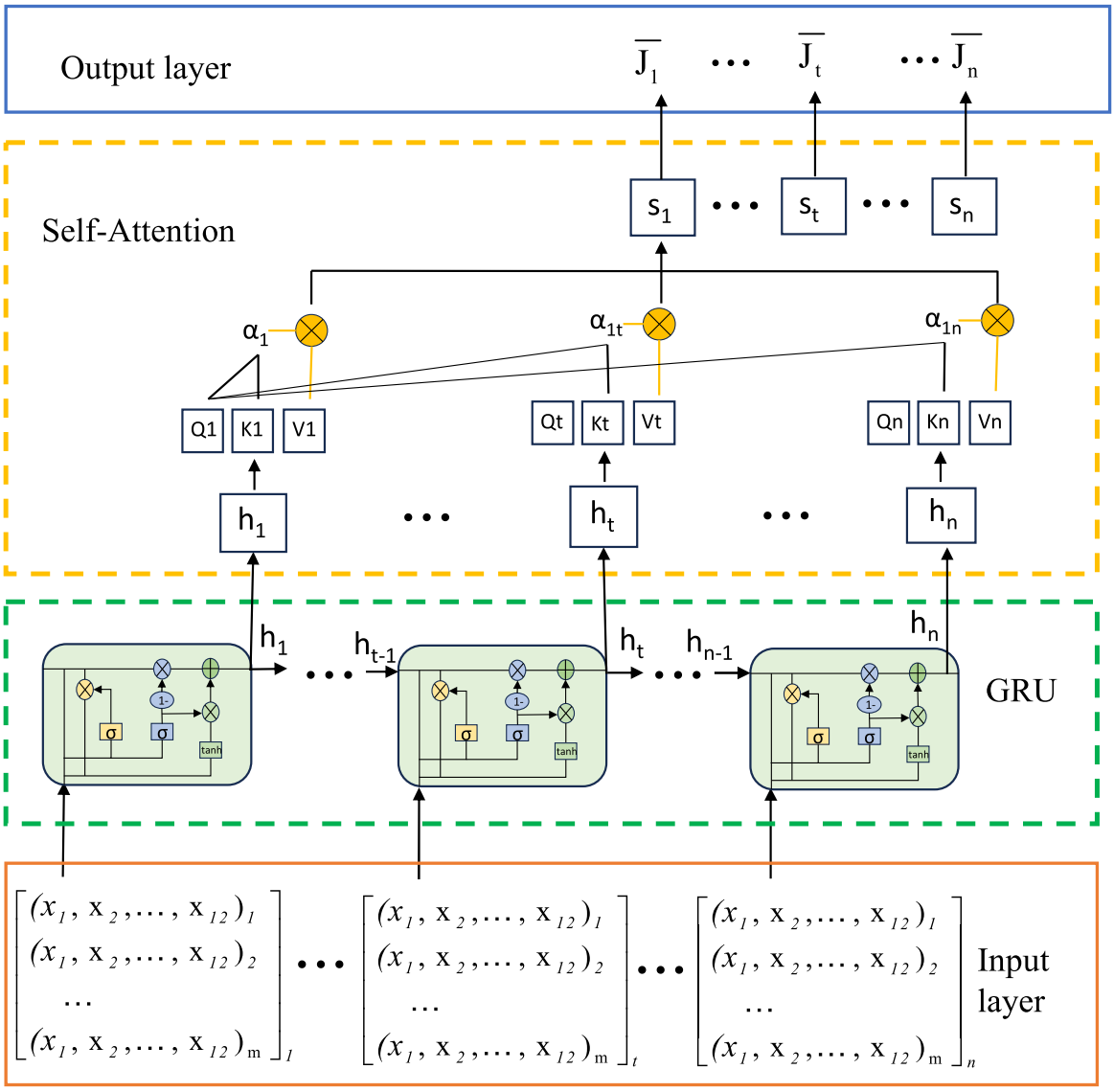


Fig. 1. Model architecture.

network that excels in modeling time series data, capable of capturing both long-term and short-term dependencies within the shrinkage and creep time series [31]. The self-attention mechanism is a resource allocation strategy that assigns different weights to the input shrinkage and creep sequences [32]. This ensures that features containing important information do not diminish over time, highlighting the influence of more significant information. Consequently, the model can more effectively learn the long-range dependencies within the shrinkage and creep sequences. The GRU and Self-Attention modules work synergistically to help the model understand key information in the shrinkage and creep sequences and generate corresponding regression predictions.

The model architecture diagram of the GRU with the self-attention mechanism model (GSA model) is shown in Fig. 1. Taking creep as an example, the internal computations during forward propagation are as follows:

1. Input Layer: The input layer takes the input feature parameters, and the input at the t-th time step is denoted as:

$$X_t = \begin{bmatrix} (t, J, type, c, w/c, a/c, V/S, f_{28}, E_{28}, RH, T, t_0)_1 \\ (t, J, type, c, w/c, a/c, V/S, f_{28}, E_{28}, RH, T, t_0)_2 \\ \cdots\cdots \\ (t, J, type, c, w/c, a/c, V/S, f_{28}, E_{28}, RH, T, t_0)_m \end{bmatrix}_t \quad (1)$$

2. GRU Layer: The hidden layer size of GRU is set to 256, with 2 layers using bidirectional GRU. The GRU layer learns feature vectors from the input feature sequence X_t, and the internal operations of the network are as follows:

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \quad (2)$$

where h_t is the output at the t-th time step, and the operations within the GRU are given by:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]) \quad (3)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t * h_{t-1}, x_t]) \quad (4)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]) \quad (5)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (6)$$

$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} \tag{7}$$

where x_t represents the input feature time series; r_z and r_t represent the reset and update gates; $\widetilde{h}_t$ is the candidate hidden state output; h_t is the current state output; W_r, W_t, and W_h are the weights for the reset gate, update gate, and candidate hidden state, respectively; h_{t-1} is the previously hidden state output.

3. Self-Attention Mechanism Layer: The input to the self-attention mechanism layer is the output vector ht from the GRU layer. This layer calculates the attention scores, weights, and the weighted sum as follows:

(1) Creating the query vector (Q), key vector (K), and value vector (V) for the GRU layer output vector h_t through linear transformations:

$$Q_t = W_Q h_t + b_Q \tag{8}$$

$$K_t = W_K h_t + b_K \tag{9}$$

$$V_t = W_V h_t + b_V \tag{10}$$

(2) Calculating attention scores: Compute the similarity between the query vector Q and the key vector K, which determines the importance of each calculation age's feature vector relative to others. For each h_i, calculate its attention scores with all other time steps.

$$\text{AttentionScore}(h_i, h_j) = \frac{Q_i \cdot K_j}{\sqrt{d}} \tag{11}$$

(3) Calculating attention weights: Use the softmax function to compute the normalized attention weights, converting the similarities into a probability distribution.

$$\text{AttentionWeight}(h_i, h_j) = \text{Softmax}\left[\text{AttentionScore}(h_i, h_j)\right] \tag{12}$$

Where:

$$\text{Softmax}(x) = \frac{\exp(x)}{\sum_{i=1}^{n} \exp(x)} \tag{13}$$

(4) Calculating the weighted sum: Apply the self-attention weights to perform a weighted sum of all elements' values (V) to obtain the output s_t for each time step. Finally, perform batch computation on the entire sequence to obtain the final output matrix.

$$s_t = \sum_{j=1}^{n} \text{AttentionWeight}(h_i,\ h_j) * V_j \tag{14}$$

4. Output Layer: The output layer receives the output s_t from the self-attention layer and generates the final regression prediction.

$$\overline{J_t} = W_{fc} s_t + b_{fc} \tag{15}$$

2.3. Data preparation

The NU database for concrete creep and shrinkage [17], established by Professor Bazant at Northwestern University, is used. As of 2021, the database has collected 1439 sets of creep tests and 3689 sets of shrinkage tests.

Due to the variations in parameter categories and experiment durations among different test groups in the NU database, appropriate feature engineering can enhance the predictive performance of the deep model. Therefore, it is necessary to preprocess the database data, with the specific steps as follows:

Step 1: Data cleaning: Remove any missing values from the input parameters.

Step 2: Selection of test groups with sufficient sequence length: Since the development of shrinkage and creep exhibits significant time accumulation, the time series for shrinkage and creep must be long enough to have predictive value. Therefore, test groups with more than five recorded instances was selected.

Table 2
Numerical encoding for cement types.

Cement type	SL	N	R	RS
Coding number	1	2	3	4

Note: SL, N, R, and RS represent slow-hardening cement, normal cement, rapid-hardening cement, and rapid-hardening high-strength cement, respectively.

Step 3: Logarithmic transformation of calculation age: The development trend of concrete shrinkage and creep is characterized by rapid growth in the early stages and slower growth in the later stages. This results in a high concentration of data in the early stages and sparse data in the later stages, which can negatively impact the model's predictive accuracy. By applying a logarithmic transformation to the calculation age t, the nonlinearity of shrinkage and creep can be significantly reduced over time. This transformation helps to achieve a more uniform distribution of the shrinkage and creep time series data, thereby enhancing the model's predictive accuracy.

Step 4: Numerical encoding of non-numeric parameters: Convert non-numeric feature parameters (such as cement type) into numeric values, as shown in Table 2.

Step 5: Time interpolation processing: standardize the time intervals for each test group. Due to the large sample size in the NU database, the data recording times for each test group vary. To facilitate time series prediction, it is necessary to standardize the time intervals for each test group.

Since the logarithmically transformed shrinkage and creep sequences approximate linearity, linear interpolation is used to unify the calculation ages of each test group, as shown in Table 3. Note that the interpolation should be applied to the logarithmically transformed shrinkage and creep sequences, and the new interpolation time points should fall within the original time points' range.

Step 6: Normalization: Different parameter features have various units and ranges, which can cause the model to be disproportionately influenced by features with larger scales, thus hindering its ability to learn other features adequately. This can negatively impact the model's generalization performance.

Moreover, if the scale differences between features are substantial, the model might require more iterations to find suitable weight parameters, making it challenging for the model to converge. Therefore, data normalization is necessary. In this study, the MinMaxScaler method is used to normalize each set of feature data. The normalization formula is as follows:

$$X_i' = \frac{X_i - X_{min}}{X_{min} - X_{max}} \tag{16}$$

in which X_i represents the feature value of the original data, X_{min} represents the minimum value of the feature in the dataset, X_{max} represents the maximum value of the feature in the dataset, and X_i' represents the normalized feature value, which ranges between 0 and 1.

After the above preprocessing steps, the NU database consists of 427 sets of creep tests and 529 sets of shrinkage tests. The parameter ranges after preprocessing are shown in Table 1. More than half of the tests in the NU database lack the data for the elastic modulus E_{28}. Some studies, such as [23], have used standard formulas to estimate E_{28} based on concrete strength to supplement the data samples. However, the elastic modulus itself has substantial variability. Therefore, using standard corrections might adversely affect the model's predictive performance. Consequently, this study opts to exclude the test groups lacking parameter data.

A sliding window approach is employed for single-step predictions. Assuming the time step length is m, the model uses creep data from the previous m time steps to predict the shrinkage and creep at the current time step m+1.

After logarithmic transformation and interpolation of the time series,

Table 3
Unified time intervals for interpolation (days).

t1	t2	t2	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13	t14	t15	t16	t17
1	2	3	4	5	6	7	14	28	60	90	120	150	180	360	1080	3600

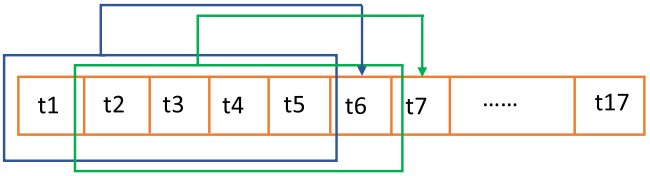


Fig. 2. The sliding window for single-step predictions.

Table 4
Sample sizes for creep and shrinkage test datasets.

Dataset	Training Set	Validation Set	Total Samples
Creep Test	2840	710	3550
Shrinkage Test	3747	937	4684

the time indices for the shrinkage and creep sequences are t1~t17, with t1~t10 being obtained from experiments. There are three future states t15~t17 that need to be predicted. The subsequent research adopts a multi-step recursive prediction strategy. To ensure the model has sufficient historical data for accurate predictions, the time step length should not be less than 5. However, if the time step length is too large, the model may fail to capture short-term fluctuations and may not have enough training samples. Therefore, m=5 is chosen as the time step length for the model in this study. The sliding window concept is illustrated in Fig. 2.

For the same set of creep tests, the data partitioning is performed as shown in Eq. 17 and 18. Within the same test group, each data set, except for the calculation age t and the historical creep function J, has other input parameters that remain constant, denoted by C.

$$\begin{bmatrix} t_1,\ J_1,\ C \\ t_2,\ J_2,\ C \\ \cdots\cdots \\ t_5,\ J_5,\ C \end{bmatrix} \rightarrow \overline{J}_6 \begin{bmatrix} t_2,\ J_2,\ C \\ t_3,\ J_3,\ C \\ \cdots\cdots \\ t_6,\ J_6,\ C \end{bmatrix} \rightarrow \overline{J}_7 \cdots\cdots \begin{bmatrix} t_{i-5},\ J_{i-5},\ C \\ t_{i-4},\ J_{i-4},\ C \\ \cdots\cdots \\ t_i\ ,\quad J_i,\quad C \end{bmatrix} \rightarrow \overline{J}_{i+1} \tag{17}$$

$$C = \left(type, c, w/c, a/c, V/S, f_{28}, E_{28}, RH, T, \mathrm{t_0}\right) \tag{18}$$

For different test groups in the NU database, the DataFrameGroupBy function from the Pandas package in Python is used to group the test groups by their respective identifiers. The data for each test group is then partitioned using the aforementioned method.

The preprocessed NU dataset was thoroughly shuffled and then divided into training and validation sets using an 80:20 ratio. The validation set consists of data that the model has never seen before. When predicting on the validation set, the model switches to evaluation mode, and gradient descent is not computed. The sample sizes for the creep test dataset and the shrinkage test dataset are shown in Table 4.

2.4. Model training

[21,22] used the LSTM for predicting shrinkage and creep, achieving excellent prediction results. For performance comparison, the LSTM is also trained on the training set in this study.

For each training sample, the model's prediction output is computed through forward propagation. To evaluate the difference between the predicted output and the actual value (label) y_i, a loss function needs to be calculated. In this study, the Mean Squared Error (MSE) is used as the loss function, as shown in Eq. (19). In this equation, n represents the number of samples, yi represents the actual value (label), and $\hat{y}_i$ represents the model output.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(\mathrm{y}_i - \hat{\mathrm{y}}_i\right)^2 \tag{19}$$

The initial learning rate was set to 1×10^{-4}, the batch size for training was set to 100, and the number of epochs was set to 100. Through backpropagation, the gradients of the loss function concerning the model parameters were computed. The Adam optimizer (adaptive moment estimation) was used to optimize and update the model weight parameters.

The training loss function of the deep network model is shown in Fig. 3. As the number of training epochs increases, the training loss values for both the GSA and LSTM models gradually stabilize. From Fig. 3(a), it can be observed that the creep model begins to converge when the training epochs reaches 20. Compared to the LSTM creep model, the GSA creep model converges faster, effectively saving training time. From Fig. 3(b), it is evident that the shrinkage model begins to converge when the number of training epochs reaches 40. The GSA

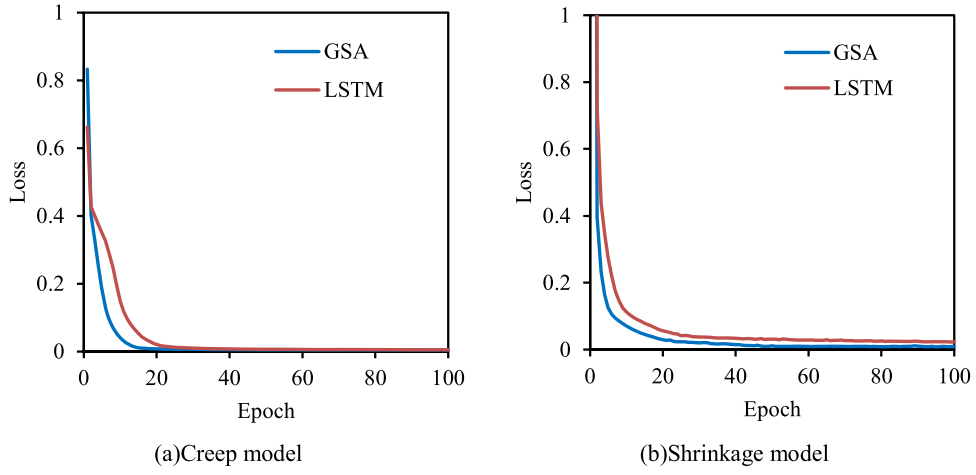


Fig. 3. The training loss function of the deep learning model.

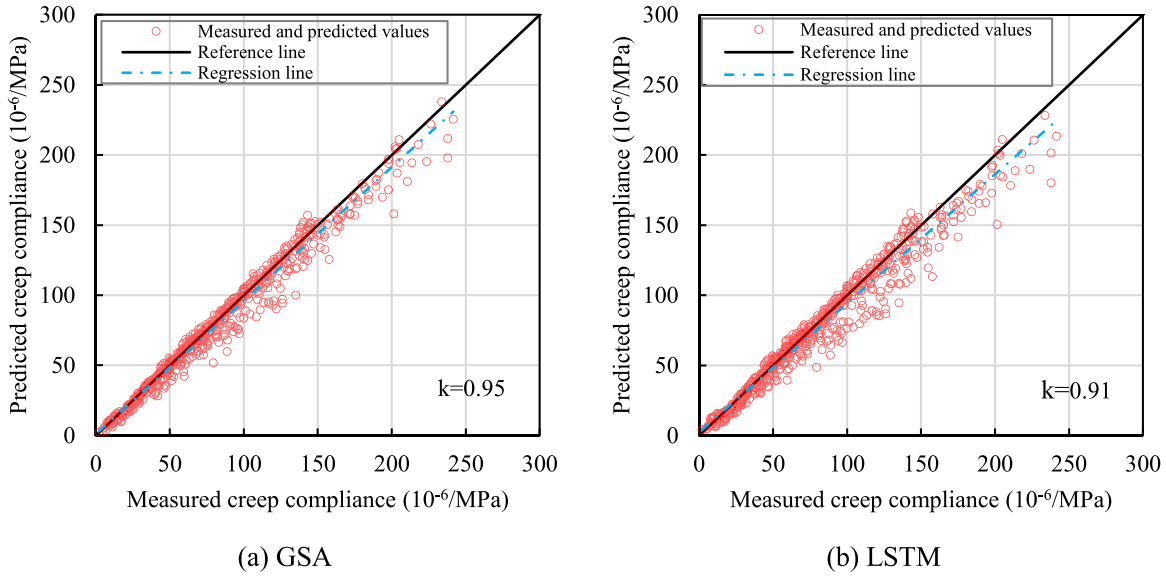


Fig. 4. The prediction results of creep compliance of the deep learning models.

shrinkage model also converges faster compared to the LSTM shrinkage model; when the number of training epochs reaches 100, the GSA shrinkage model's loss value is 0.009, which is 60 % smaller than the LSTM model's loss value of 0.023.

2.5. Model validation

(1) Performance index

Four indexes were selected to evaluate the performance of the proposed model: the correlation coefficient r, the coefficient of determination R^2, the mean absolute error (MAE), and the root mean square error (RMSE). Among them, r and R^2 measure the correlation between the independent and dependent variables, as well as the model fitting effect and parameter estimation, with closer to 1 indicating higher model accuracy. MAE and RMSE measure the differences between the predicted values and the actual values, with smaller values indicating higher accuracy. The formulas for these indexes are shown in Eqs. (20) to (23).

$$r = \frac{\sum_{i=1}^{n}(f_i - \bar{f}_i)\sum_{i=1}^{n}(\hat{f}_i - \bar{\hat{f}}_i)}{\sqrt{\sum_{i=1}^{n}(f_i - \bar{f}_i)^2 \sum_{i=1}^{n}(\hat{f}_i - \bar{\hat{f}}_i)^2}} \tag{20}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(f_i - \hat{f}_i)^2}{\sum_{i=1}^{n}(f_i - \bar{f}_i)^2} \tag{21}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{n}|f_i - \hat{f}_i| \tag{22}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(f_i - \hat{f}_i)^2} \tag{23}$$

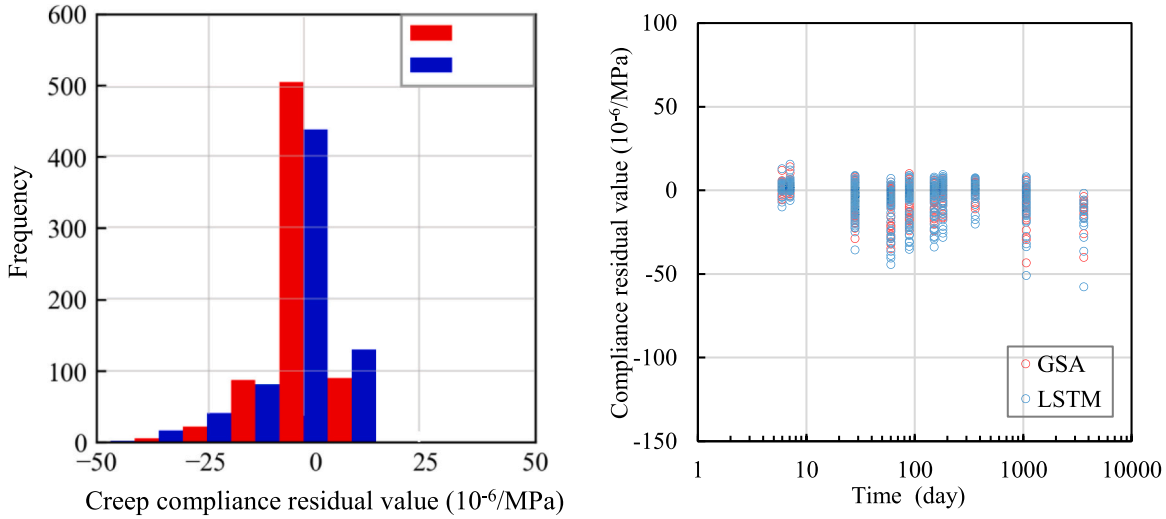


Fig. 5. Residual distribution of creep compliance of the deep learning models.

Table5
The performance indexes of model J_{LSTM} and J_{GSA}.

Model	r	R^2	MAE	RMSE
J_{LSTM}	0.982	0.956	6.340	10.152
J_{GSA}	0.989	0.975	4.771	7.691

Where f_i , $\bar{f}_i$, $\hat{f}_i$, $\bar{\hat{f}}_i$ represent the actual value, the mean of the actual values, the predicted value by the model, and the mean of the predicted values by the model, respectively.

(2) Validation of Creep Model

The scatter plots of the predicted values and actual values for the GSA and LSTM creep models on the validation set are shown in Fig. 4. The reference line represents the linear equation with a slope of 1, indicating that data points closer to this line have predicted values closer to the actual values. The regression line is obtained using the least squares method to perform linear regression on the data points, with k representing the slope of the regression equation.

From Fig. 4, it can be seen that most data points are distributed along the reference line y=x, with only a few data points deviating. This indicates that the predicted values are very close to the actual values. As shown in Fig. 4(a) and (b), the slopes k of the regression lines for the GSA and LSTM models are 0.95 and 0.91, respectively, indicating that the GSA model has better predictive accuracy.

Fig. 5 shows the residual analysis results for the GSA and LSTM creep models on the validation set. Residuals represent the differences between the predicted and actual values. From Fig. 5(a), it is evident that the residuals of the GSA and LSTM creep models mainly fall between -15×10^{-6}/MPa and 10×10^{-6}/MPa, but the residuals of the GSA creep model are more concentrated and closer to 0. Fig. 5(b) displays the residuals over time, showing that the GSA model's residuals are smaller than those of the LSTM model most of the time, indicating that the GSA model has superior predictive capability.

Similarly, the evaluation metrics in Table 5 also indicate that the GSA creep model has better predictive accuracy.

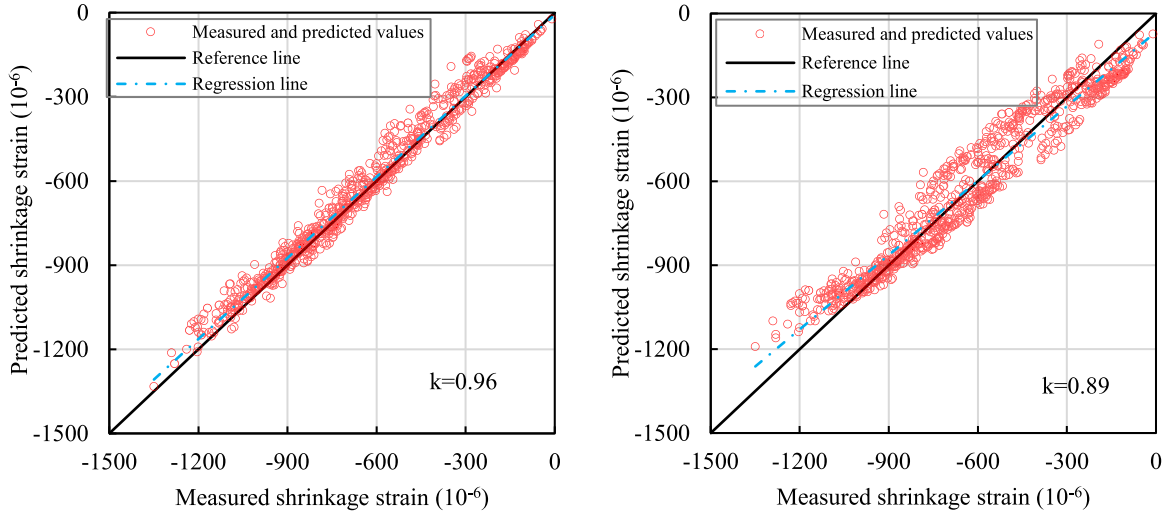


Fig. 6. The prediction results of shrinkage strain of the deep learning models.

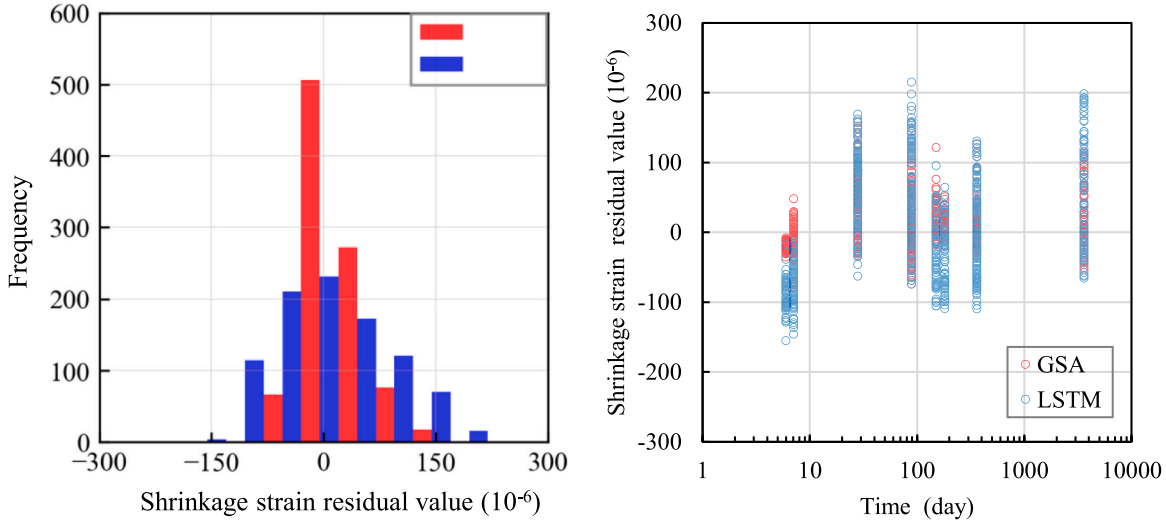


Fig. 7. Residual distribution of shrinkage strain of the deep learning models.

Table 6
The performance indexes of model ε_{LSTM} and ε_{GSA}.

Model	r	R^2	MAE	RMSE
ε_{LSTM}	0.971	0.939	62.342	75.741
ε_{GSA}	0.992	0.980	32.194	43.254

(3) Validation of Shrinkage Model

The scatter plots of the predicted and actual values for the GSA and LSTM shrinkage models on the validation set are shown in Fig. 6. From Fig. 6, it can be seen that most data points are distributed along the reference line y=x, with only a few data points deviating. This indicates that the predicted values are very close to the actual values. Fig. 6(a) and (b) show that the slopes k of the regression lines for the GSA and LSTM models are 0.96 and 0.89, respectively, indicating that the GSA shrinkage model has better predictive accuracy.

Fig. 7 shows the residual analysis results for the GSA and LSTM shrinkage models on the validation set. From Fig. 7(a), it is evident that the residuals of the GSA shrinkage model mainly fall between -120×10^{-6} and 120×10^{-6}, whereas the residuals of the LSTM shrinkage model range from -120×10^{-6} and 150×10^{-6}. This indicates that the residuals of the GSA shrinkage model are more concentrated and closer to 0. Fig. 7(b) displays the residuals over time, showing that the GSA model's residuals are smaller than those of the LSTM model most of the time, indicating that the GSA shrinkage model has superior predictive capability.

Similarly, the evaluation metrics in Table 6 also indicate that the GSA shrinkage model has better predictive accuracy.

3. Shrinkage and creep tests

3.1. Test overview

This experiment was conducted as part of the Lugou River Bridge project, a multi-span continuous rigid-frame bridge with ultra-high piers located in the mountainous region of Southwest China. The geographical location is shown in Fig. 8, situated in Chongqing. The region experiences a subtropical humid monsoon climate, characterized by mild temperatures and abundant rainfall, with an average annual relative humidity of 79 % and an average temperature of 17.72°C.

Considering the experimental environment and loading age, a total of 8 groups of shrinkage and creep tests were designed, as shown in Table 7.

The concrete mix ratio used for the experiment follows the C60 concrete of the Lugou River Bridge main girder, as shown in Table 8. According to the standard GB/T 50082–2009 [33], the shrinkage and creep tests used prismatic specimens of 100 mm×100 mm×300 mm. Three specimens were made for each test condition. To determine the initial loading force for the creep test, three additional prismatic specimens of the same size were made simultaneously with each batch of shrinkage and creep test specimens.

According to the standard GB/T 50081–2019 [34], cubic strength specimens for the concrete material properties test were made using

Table 7
Test conditions for shrinkage and creep experiments.

No.	Test number	Type	Environment	Loading age (day)
1	COC3	Creep	Outdoor	3
2	COC7	Creep	Outdoor	7
3	CLC3	Creep	Indoor	3
4	CLC7	Creep	Indoor	7
5	SOC3	Shrinkage	Outdoor	3
6	SOC7	Shrinkage	Outdoor	7
7	SLC3	Shrinkage	Indoor	3
8	SLC7	Shrinkage	Indoor	7

Table8
Mix ratio for the C60 concrete.

Component	Amount (kg/m^3)
Cement (P.O 52.5)	491
Coarse aggregate	700
Fine aggregate (5–10)mm	210
Fine aggregate (10–20)mm	841
Water	153
Fly ash	55
Water reducer	8.19

Fig. 8. The location of the field test.

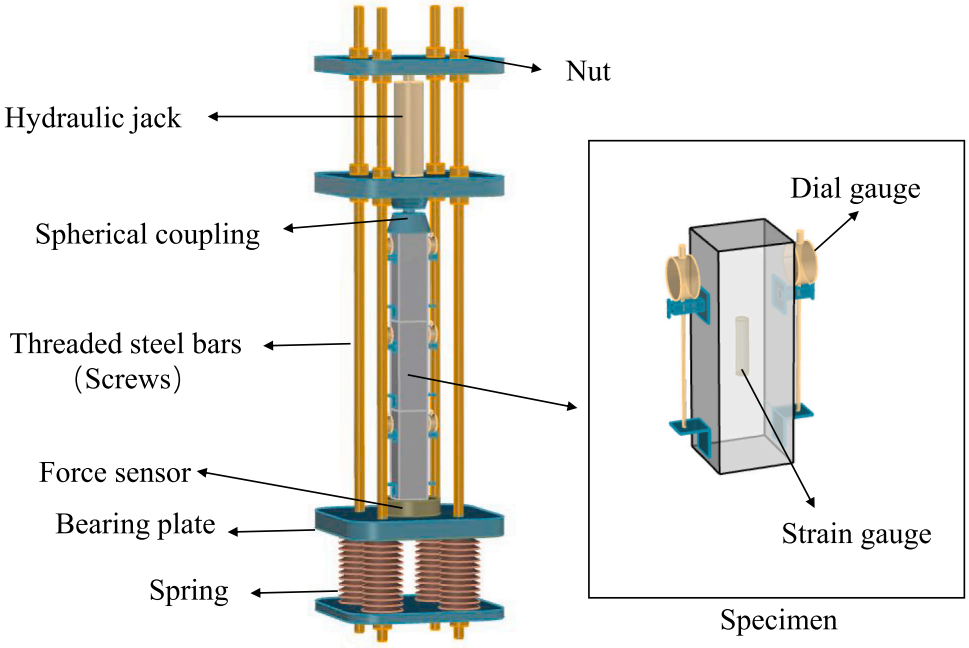


Fig. 9. Schematic diagram of the spring-type creep loading instrument (cm).

150 mm×150 mm×150 mm cubes, while prismatic specimens for elasticity modulus tests and other properties used 100 mm×100 mm×300 mm prisms. Three specimens are made for each test condition.

The outdoor shrinkage and creep specimens were cured using the same method as the actual bridge construction segments. After demolding, the specimens were placed at the construction site at the base of the bridge piers, covered with geotextile, and regularly watered for curing. The indoor shrinkage and creep specimens, as well as the concrete material property specimens, were cured under standard conditions. After demolding, these specimens were placed in a standard curing room with a temperature of 20±2°C and a relative humidity of over 95 %.

3.2. Loading and measurement scheme

The creep test in this study used a custom-made spring-loaded creep-loading device, as shown in Fig. 9. The creep loading device consists of four threaded rods arranged in a square layout to ensure even force distribution. The threaded rods are made of precision-rolled threaded steel with a diameter of 32 mm, each having an ultimate bearing capacity of 26.5 tons. The device includes four pressure plates, with both the upper and lower plates having a thickness of 4 cm, and both the top and bottom plates also having a thickness of 4 cm.

Between the lower pressure plate and the bottom plate, there are four sets of disc springs. Each set of disc springs has an outer diameter of 140 mm, an inner diameter of 72 mm, a total height of 145.6 mm, and a stroke of 20 mm when loaded with 6 tons. When the threaded rods are subjected to 30 % of their ultimate bearing capacity, the device's bearing capacity is approximately 30 tons, meeting the requirements for creep tests as specified in the standard [33].

Considering the complexity of the onsite construction environment, relying solely on dial gauges for measurements may lead to potential damage. Therefore, to ensure data reliability, both dial gauges and

(a)Outdoor test (b)Indoor test

Fig. 10. Shrinkage and creep field tests.

Table 9
The prismatic strength results of the shrinkage and creep specimens (MPa).

	COC7/SOC7	COC3/SOC3	CLC7/SLC7	CLC3/SLC3
Test	46.7	48.2	56.9	47.4
28-day fitted values	57.1	72.7	69.5	71.5

embedded strain sensors were used. The dial gauges measure the surface strain of the specimens, while the embedded strain sensors measure the internal strain. The final strain value of each specimen is taken as the average of the internal and external strains. The arrangement of the strain measurement instruments is shown in Fig. 9.

The dial gauges were manufactured by Liangju Renju Company, with a range of 1 mm and an accuracy of 0.001 mm. The embedded strain sensors were used JM215 series strain sensors, produced by Sichuan Jinma Company, with a range of ±1500 με and an accuracy of 1 με, suitable for operating temperatures of 10°C to 70°C. Besides measuring strain, the JM215 sensor can also measure the internal temperature of the concrete.

The outdoor shrinkage and creep tests were conducted on the bridge construction site, from April 2023 to October 2023. The indoor tests were carried out in the constant temperature and humidity laboratory, from May 2023 to November 2023. The test site is shown in Fig. 10.

According to the standard GB/T 50082–2009[33], the stress level for the creep loading in this test is set at 0.4. This means that the creep loading stress is 40 % of the compressive strength of the prismatic specimens at the corresponding age. The creep loading process follows the standard [33]: first, the load is applied to 20 % of the creep load, then alignment adjustments are made continuously. Once completed, the load is increased to the creep load, the corresponding nuts are tightened, and the jack is removed.

3.3. Test result

The prismatic strength results of the shrinkage and creep specimens are shown in Table 9, where the 28-day fitted values are calculated according to the fib2010 standard. The recorded ambient temperature and relative humidity for both outdoor and indoor tests are shown in Fig. 11. The environmental temperature and relative humidity are taken as the average values during the test period, with indoor and outdoor average temperatures of 21.2°C and 25.6°C, respectively, and indoor and outdoor average relative humidity of 53.9 % and 79.6 %, respectively. Overall, the outdoor tests had higher average ambient temperature and relative humidity compared to the indoor tests, with greater and more frequent fluctuations.

The strain measurement value of the creep and shrinkage specimens were the average of the dial indicators and the strain sensor. The creep compliance J was calculated from Eq. (24). The shrinkage strain ε_s is obtained by subtracting the thermal strain from the measured strain of the unloaded specimen, as shown in Eq. (25), where the measured strains from outdoor specimens contained thermal strains due to the significant temperature change.

$$J\left(t, t_0\right) = \frac{\varepsilon(t) - \varepsilon_{cn}(t)}{\sigma} \tag{24}$$

$$\varepsilon_s\left(t, t_s\right) = \begin{cases} \varepsilon_{cn}(t) - \alpha_c\left(T_{current} - T_{initial}\right), & Outdoor \\ \varepsilon_{cn}(t), & Indoor \end{cases} \tag{25}$$

Where ε(t) indicates the strain measured in the loaded specimen when the concrete age is t, ε_{cn}(t) indicates the strain measured in the unloaded specimen when the concrete age is t. σ indicates the loading stress of the creep test. α_T=10×10^{-6}/℃ is the coefficient of thermal expansion of concrete. $T_{current}$ is the ambient temperature at the time of measurement, and $T_{initial}$ is the ambient temperature at the time of creep loading.

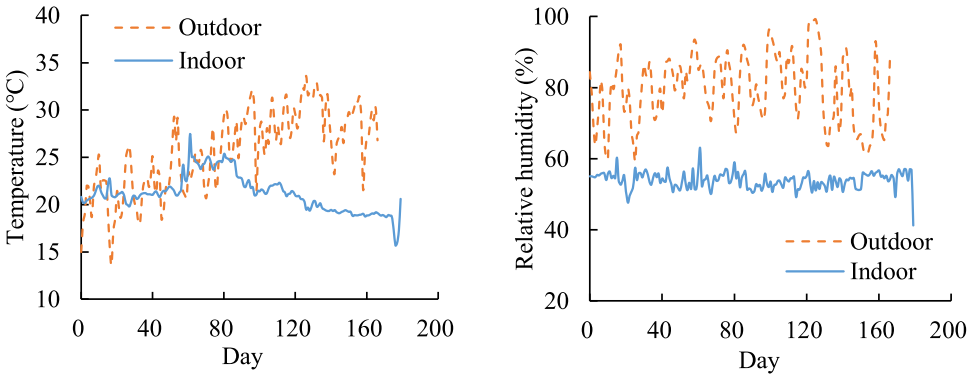
Fig. 12 presents the test results of shrinkage and creep under different environmental conditions and loading ages. As shown in Fig. 12 (a) and (b), the creep compliance and shrinkage strain in the indoor environment exhibited a stable growth trend, while the outdoor environment showed significant fluctuations due to varying temperature and humidity, as confirmed by Fig. 11. At a loading age of 180 days, the creep compliance and shrinkage in the outdoor environment were 10 % and 400 % lower than those in the indoor environment, respectively. Fig. 12 (c) and (d) indicate that smaller loading ages result in higher creep coefficients and shrinkage strains in the concrete.

To obtain the historical parameters of shrinkage and creep for the deep learning model, the field-measured data from the shrinkage and creep tests were logarithmically transformed and linearly interpolated to obtain creep functions and shrinkage strain values at uniform time intervals, as shown in Table 10 and Table 11.

4. Short-term prediction

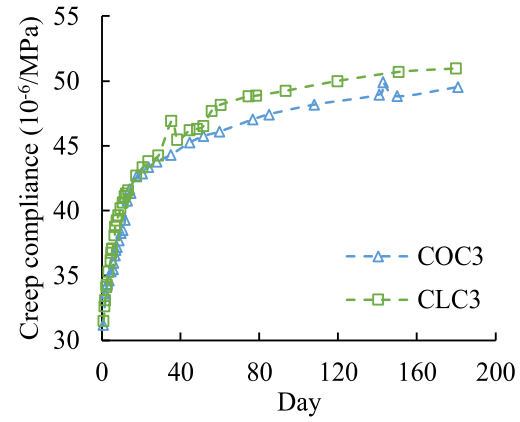
4.1. Comparison of the model prediction with the creep tests

The basic parameters of the shrinkage and creep tests were input into the trained GSA model for single-step predictions. Additionally, the fib2010 [35], B4 [6], AASHTO [9], ACI [36], and GL2000 [37] models were used to calculate and compare each test group's results. The coefficient of variation V was used to evaluate the predictive performance of each model. The calculation formulas for the coefficient of variation are shown in Eq. (26) to Eq. (28) [38]. The essence of the coefficient of variation is the root mean square error (RMSE), with smaller values indicating better predictive performance of the model.

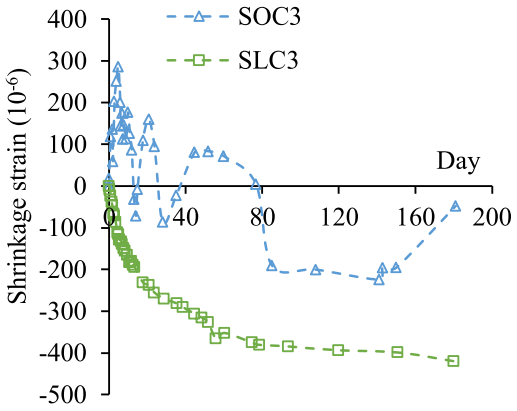


(b) Temperature (b) Relative humidity

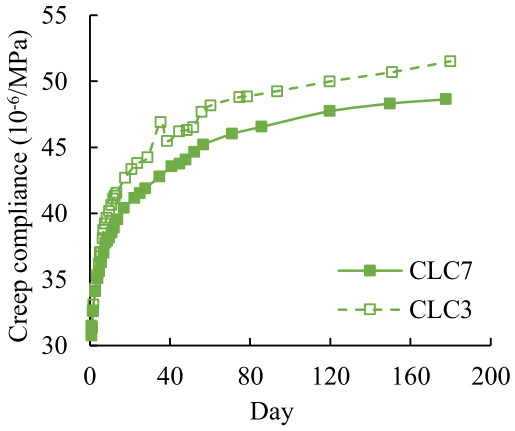
Fig. 11. Daily temperature and relative humidity.



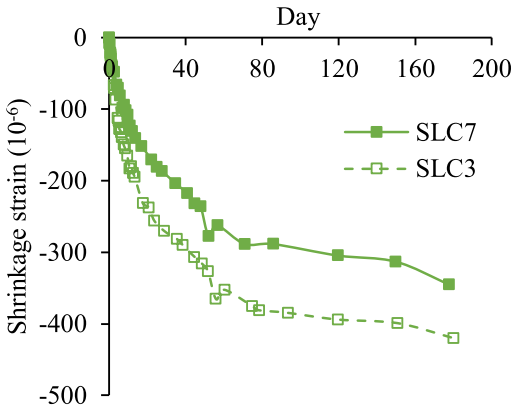
(a) Test results of creep in different environments



(b) Test results of shrinkage in different environments



(c) Test results of creep at different loading ages



(d) Test results of shrinkage at different loading ages

Fig. 12. Test results of shrinkage and creep.

Table 10
Creep compliance values from the field tests (10^{-6}/MPa).

	t1	t2	t2	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13	t14
COC7	31.1	36.2	40.2	44.0	46.6	48.6	49.0	56.1	58.7	61.5	64.4	66.8	68.3	69.8
COC3	27.8	31.1	33.0	34.1	34.5	34.9	35.2	39.6	42.4	44.7	47.1	49.4	47.9	48.6
CLC7	27.7	30.9	33.2	34.4	35.2	35.8	36.5	39.3	41.8	45.4	46.7	47.7	48.3	48.8
CLC3	-	31.7	33.5	34.7	35.8	36.8	37.7	41.5	44.1	48.1	49.1	50.0	50.7	51.5

Table 11
Shrinkage strain values from the field tests (10^{-6}).

	t1	t2	t2	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13	t14
SOC7	−3	−63	−92	−136	−86	−88	−99	−178	−240	−222	−272	−326	−378	−287
SOC3	20	39	15	74	95	102	56	−47	−72	−37	−133	−145	−144	−200
SLC7	−12	−27	−43	−52	−66	−74	−85	−137	−185	−268	−290	−304	−314	−345
SLC3	-	−26	−54	−77	−101	−118	−131	−193	−266	−356	−384	−394	−399	−432

$$\overline{O_j} = \frac{1}{n}\sum_{j=1}^{n} O_j \tag{26}$$

$$V = \frac{1}{\overline{O_j}}\sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(C_j - O_j\right)^2} \tag{27}$$

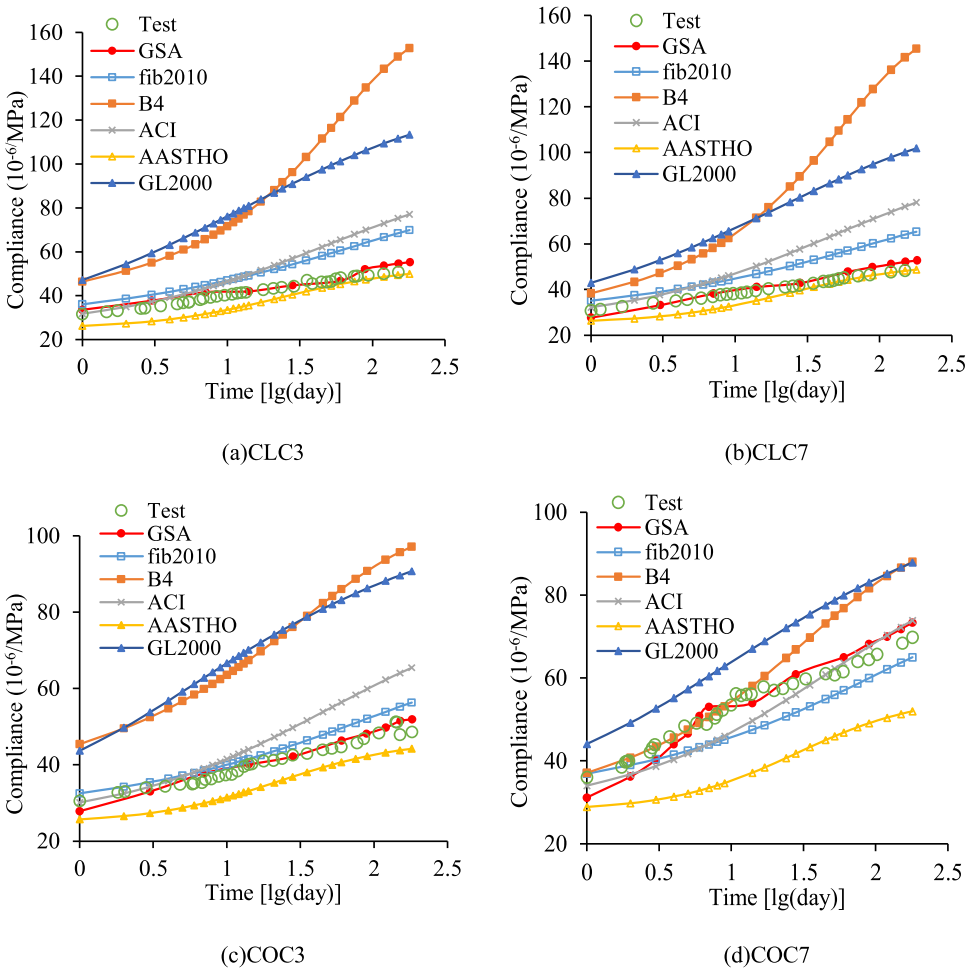


Fig. 13. Comparison of experimental creep compliance and model predictions.

$$V_{CEB} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} V_i^2} \tag{28}$$

Where V represents the coefficient of variation for a single test group with the model, V_{CEB} represents the coefficient of variation for all tests with the model (indicating overall variability), n represents the total number of data points, N represents the total number of test groups, O_j represents the measured value of the creep coefficient or shrinkage strain, and C_j represents the predicted value of the creep coefficient or shrinkage strain.

The comparison results between the predicted values from various models and the experimental values of the creep compliance are shown in Fig. 13. As observed in Fig. 13, the creep compliance of the outdoor specimens exhibits significant fluctuations due to the influence of environmental temperature and humidity. Among the models, only the GSA model demonstrates a similar fluctuation trend in the creep and shrinkage curves. This indicates that the GSA model can capture real-time variation trends through historical creep data, thus effectively considering the impact of fluctuating outdoor temperature and humidity.

Comparing the predictive results of the B4 model and the GSA model reveals that the B4 model consistently overestimates the creep function across all test groups, while the GSA model achieves the best predictive performance for the creep test values among all models. Since both the B4 and GSA models are based on the NU database and possess similar empirical statistical characteristics, the difference in predictive performance is likely due to the GSA model treating creep as a time series and incorporating its historical features as input. This suggests that conducting short-term tests to obtain historical characteristics is crucial for accurate creep predictions, enhancing predictive accuracy.

Table 12
The coefficients of variation for different creep models.

	fib2010	B4	ACI	AASTHO	GL2000	GSA
COC7	0.10	0.19	0.09	0.29	0.25	0.05
COC3	0.10	0.82	0.22	0.14	0.78	0.05
CLC7	0.25	1.35	0.41	0.10	0.89	0.07
CLC3	0.26	1.39	0.33	0.12	1.03	0.06
V_{CEB}	0.19	1.05	0.29	0.18	0.79	0.06

The coefficients of variation for different models in the on-site creep tests are shown in Table 12. As seen in Table 12, the GSA model consistently has the smallest coefficient of variation (V_{CEB}) for all four creep test groups, each being 0.06, indicating that the GSA model provides the most accurate predictions for the on-site creep test results. Among other models, the ranking of the coefficients of variation for the creep test results from smallest to largest is AASHTO, fib2010, ACI, GL2000, and B4.

4.2. *Comparison of the model prediction with the shrinkage tests*

The comparison results between the predicted values from various models and the experimental values of the shrinkage strain are shown in Fig. 14. As observed in Fig. 14, the shrinkage strain of the outdoor specimens exhibits significant dispersion due to the influence of

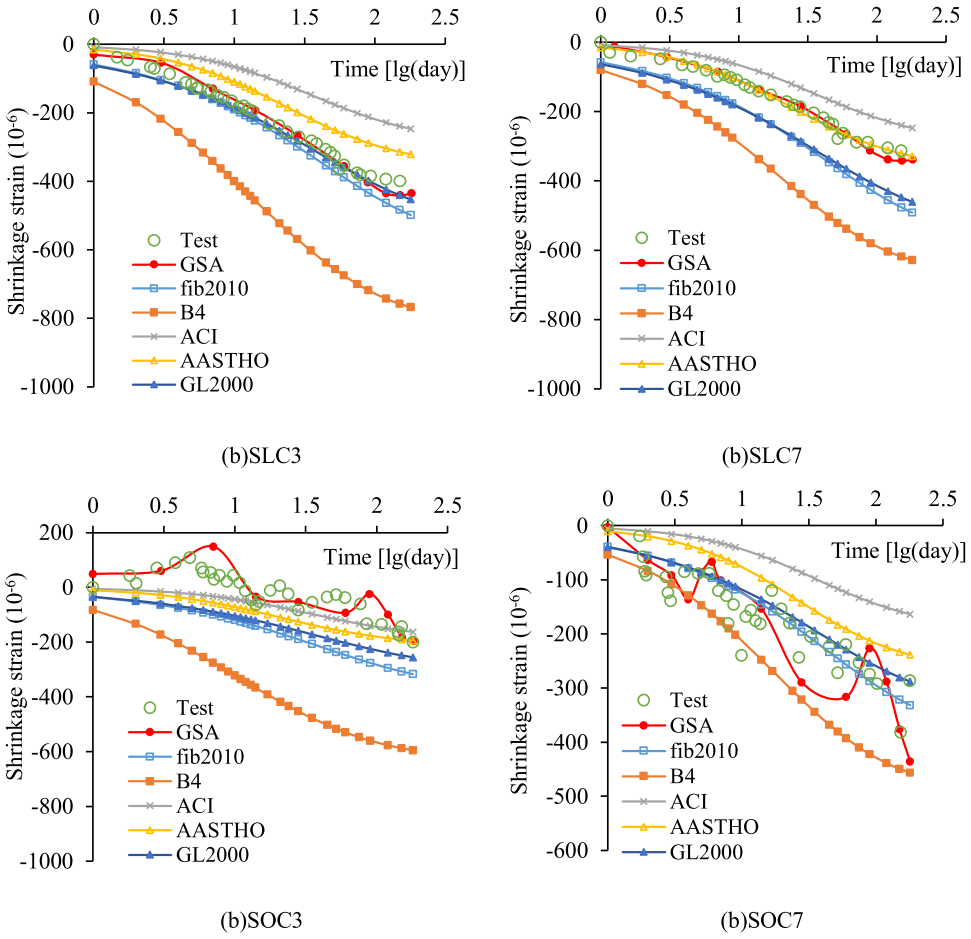


Fig. 14. Comparison of experimental shrinkage strain and model predictions.

Table 13
The coefficients of variation for different shrinkage models.

	fib2010	B4	ACI	AASTHO	GL2000	GSA
SOC7	0.24	0.54	0.66	0.44	0.26	0.31
SOC3	7.68	18.49	3.50	4.55	6.34	1.17
SLC7	0.59	1.26	0.36	0.12	0.52	0.07
SLC3	0.19	1.15	0.55	0.38	0.11	0.07
V_{CEB}	3.85	9.29	1.81	2.29	3.18	0.61

environmental temperature and humidity. Among the models, only the GSA model demonstrates a similar fluctuation trend in the predicted and experimental shrinkage strain values. This indicates that the GSA model can capture real-time variation trends through historical shrinkage data, effectively considering the impact of fluctuating outdoor temperature and humidity.

The coefficients of variation for different models in the on-site shrinkage tests are shown in Table 13. As seen in Table 13, the GSA model consistently has the smallest coefficient of variation (V_{CEB}) for all four shrinkage test groups, with a value of 0.61, indicating that the GSA model provides the most accurate predictions for the on-site shrinkage test results. Among other models, the ranking of the coefficients of variation for the shrinkage test results from smallest to largest is ACI, GL2000, AASHTO, fib2010, and B4. If the uncertain impact of fluctuating outdoor temperature and humidity on shrinkage is considered, and the two outdoor shrinkage tests are excluded, the models with better predictive performance are AASHTO and GL2000.

Table 14
Time series intervals for shrinkage and creep predictions.

Time number	t-t0(day)	lg[(t-t0)day]	Note
t1	1	0.00	Single-step Prediction
t2	2	0.30	
t3	3	0.48	
t4	4	0.60	
t5	5	0.70	
t6	6	0.78	
t7	7	0.85	
t8	14	1.15	
t9	28	1.45	
t10	60	1.78	
t11	90	1.95	
t12	120	2.08	
t13	150	2.18	
t14	180	2.26	
t15	360	2.56	Recursive Multi-step Prediction
t16	1080	3.03	
t17	3600	3.56	

5. Long-term prediction

The objective of this section is to extrapolate trends for the next ten years based on six months of experiment data. After preprocessing with linear interpolation, the time intervals for the shrinkage and creep sequences were standardized into a time series of 17 time points, as shown in Table 14. The data from t1 to t14 have been obtained through the field tests. The goal now is to predict the states at t15, t16, and t17 based on the data from t1 to t14.

As previously discussed, shrinkage and creep are typical time series

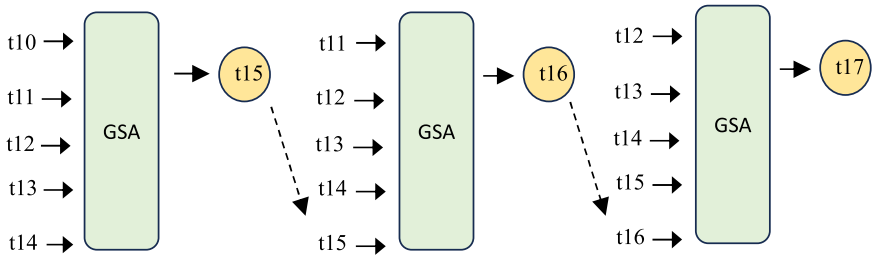


Fig. 15. Schematic diagram of the recursive multi-step prediction.

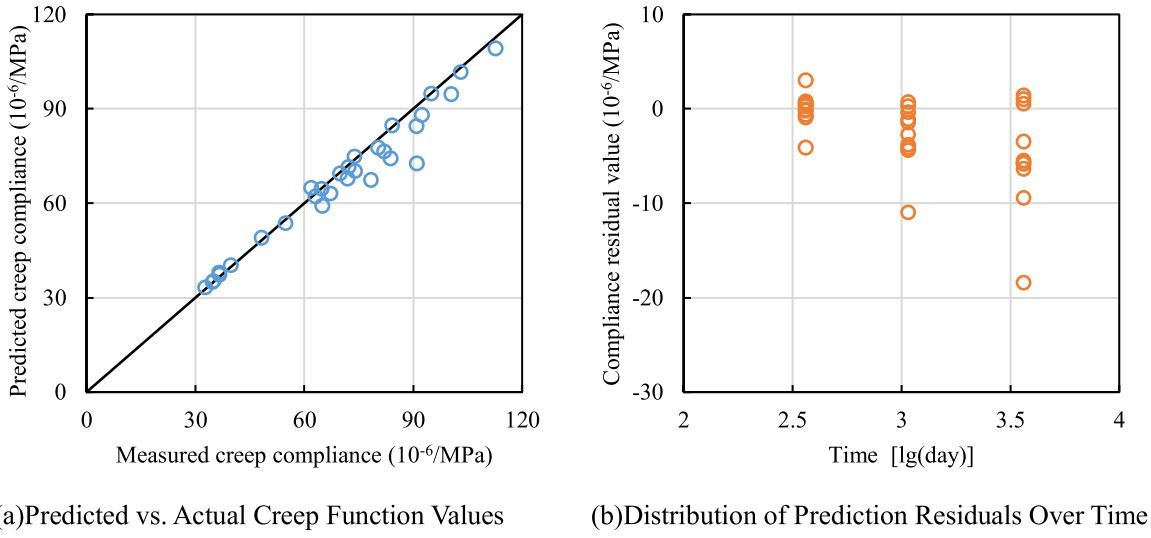


Fig. 16. Validation of recursive multi-step prediction method.

phenomena. Time series prediction can be categorized into single-step and multi-step predictions based on the length of the prediction time step. The method described earlier, which uses the previous five time points to predict the state at the sixth time point, is a single-step prediction. In this section, we need to predict the states at t15 to t17 in one go, which falls under multi-step prediction.

5.1. *Multi-step time series prediction method*

Considering the feasibility and practicality of the methods, this study employs the recursive multi-step prediction strategy for long-term shrinkage and creep prediction. This strategy uses a single-step prediction model iteratively to obtain multi-step predictions. First, the single-step model predicts the value at the first time point, then uses this predicted value as input to predict the next time point, and so on. This method simplifies model complexity by requiring only one single-step prediction model [39,40].

The GSA model is first used to perform single-step predictions sequentially for t1 to t14. Then, t10 to t14 are used to predict t15 (since the time step length is set to 5). The window slides and the newly predicted t15 is used to predict the state at t16, and so on, until t17 is predicted. The schematic diagram for recursive multi-step prediction is shown in Fig. 15.

5.2. *Validation of recursive multi-step prediction method*

To verify the accuracy of the recursive multi-step prediction method for long-term predictions, creep tests from the NU database with test durations exceeding 10 years were selected as the validation group. The recursive multi-step method was used to perform the predictions. The results are shown in Fig. 16, where the predicted results are the creep compliance for t15, t16, and t17.

From Fig. 16(a), it can be seen that most of the predicted results are distributed along the diagonal, indicating that the predicted values are very close to the actual values. The coefficient of determination R^2 calculated using Eq. (21) is 0.945, demonstrating that the GSA model with the recursive multi-step prediction method provides good accuracy for long-term creep predictions.

Fig. 16(b) shows the distribution of residuals over time for the recursive multi-step prediction method. It can be observed that most of the residuals fall within the range of -10×10^{-6}/MPa to 5×10^{-6}/MPa, indicating a slight underestimation of the actual values. The prediction error increases over time due to the randomness of long-term concrete creep and iterative errors. However, overall, the prediction errors are relatively small and within an acceptable range for engineering applications.

These results confirm that the recursive multi-step prediction method based on the GSA model effectively captures the long-term behavior of concrete creep, making it a reliable tool for long-term predictions in engineering projects.

5.3. *Long-term prediction results of shrinkage and creep*

Based on six months of shrinkage and creep test data, the GSA model and the recursive multi-step method were used to predict the development trends of shrinkage and creep over the next ten years. Two outdoor creep test groups were selected as representative creep test groups due to the highly unstable nature of outdoor shrinkage curves influenced by fluctuating temperature and humidity. Additionally, two indoor shrinkage test groups were selected as representative shrinkage test groups.

To validate the reasonableness of the long-term prediction results of

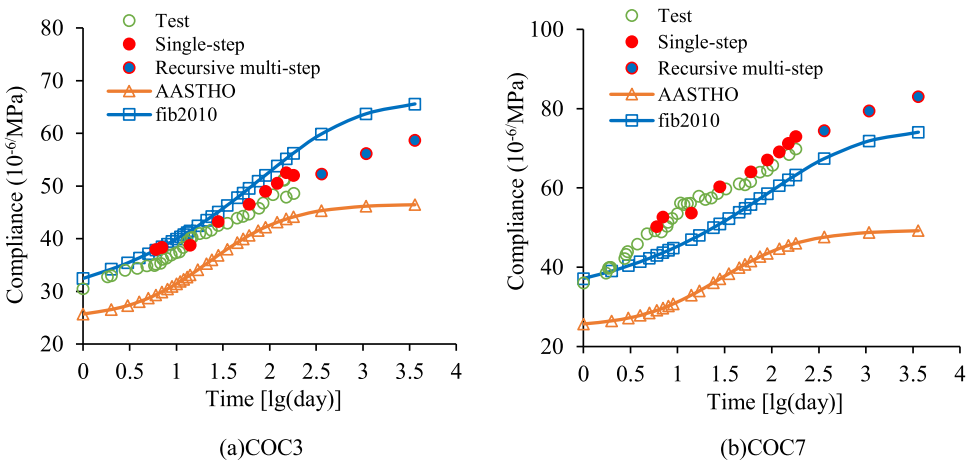


Fig. 17. Long-term prediction results of creep compliance.

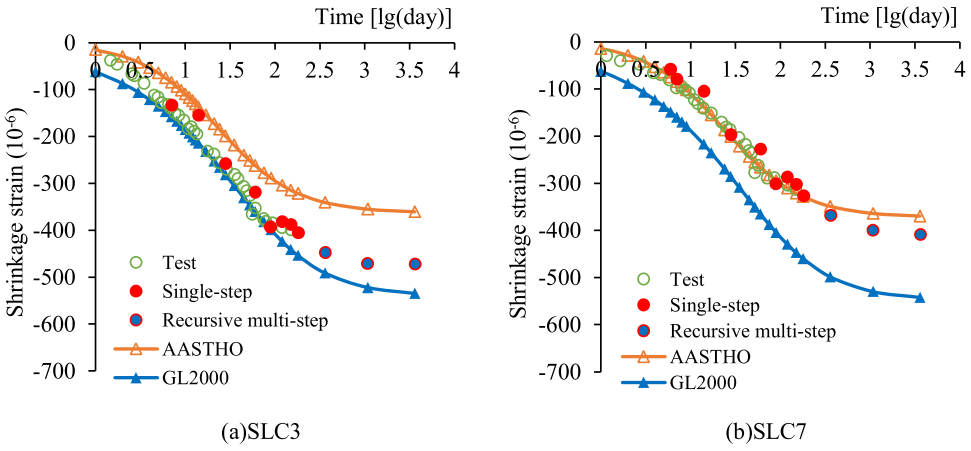


Fig. 18. Long-term prediction results of shrinkage strain.

the GSA model, comparisons were made with the AASHTO model, fib2010 model, and GL2000 model, which performed well in the short-term prediction tests.

The long-term prediction results is shown in Fig. 17 and Fig. 18. In the figures, the green, red, and blue data points represent the experimental values, GSA single-step prediction values, and GSA recursive multi-step prediction values, respectively. The GSA single-step predictions are based on data within the first six months, where all inputs to the model come from real experimental data. In contrast, the GSA recursive multi-step prediction model uses inputs that include predicted values from the previous time step, leading to an increasing proportion of predicted values as inputs the further the prediction goes.

From Fig. 17(a), it is evident that for the COC3 short-term creep test, the GSA model's single-step prediction results are closest to the experimental creep values. The AASHTO model underestimates the COC3 creep values, while the fib2010 model slightly overestimates them. For long-term creep development, the GSA recursive multi-step prediction results lie between the other two standard models, but with a higher growth rate. The AASHTO model's prediction curve stabilizes quickly, becoming nearly flat after one year. After ten years, the GSA predicted value for COC3 is 58.7×10^{-6}/MPa, an increase of 12.8 % from 52.0×10^{-6}/MPa at six months.

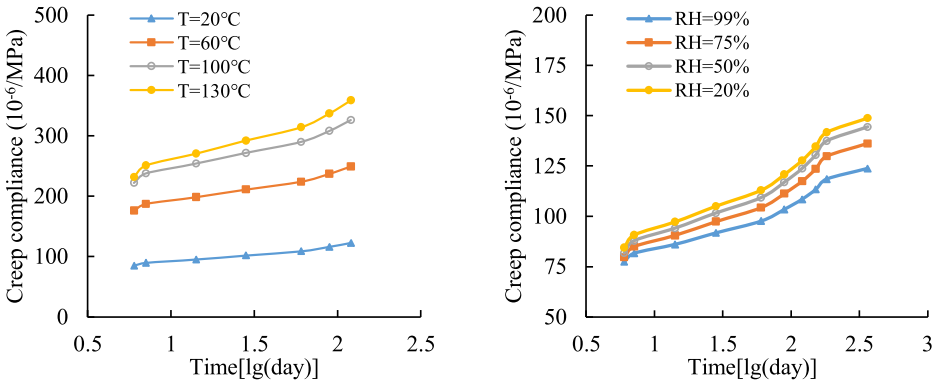
Fig. 17(b) shows the prediction results for the COC7 creep values. Unlike COC3, both the AASHTO and fib2010 models underestimate the COC7 creep values, while the GSA model's single-step prediction results are closest to the experimental values. The higher creep values of COC7 compared to COC3 may be due to differences in the concrete strength. After ten years, the GSA predicted value for COC7 reaches 83.0×10^{-6}/MPa, an increase of 13.6 % from 72.9×10^{-6}/MPa at six months.

Overall, the prediction results for the two outdoor creep test groups indicate that the GSA model has better adaptability and generalization capabilities for the random development of creep caused by variability in concrete parameters.

Fig. 18 shows the long-term prediction results for the shrinkage strain. As seen in the figures, the AASHTO model underestimates the SLC3 experimental values, while the GL2000 model overestimates the SLC7 experimental values. Only the GSA model can accurately predict both the SLC3 and SLC7 experimental values, further validating the superior accuracy of the GSA model. The recursive multi-step predictions of the GSA model fall between the AASHTO and GL2000 models and tend to align more closely with traditional models that better match short-term experimental values, indicating that the long-term predictions of the GSA model are reasonable.

After ten years, the GSA predicted value for SLC3 is -472×10^{-6}, an increase of 16.5 % from -405×10^{-6} at six months. For SLC7, the GSA predicted value is -409×10^{-6} after ten years, an increase of 25.4 % from -326×10^{-6} at six months.

The results demonstrate that the GSA model not only accurately captures the short-term behavior of shrinkage and creep but also provides reliable long-term predictions. The model's ability to utilize historical data for multi-step predictions makes it a robust tool for assessing the long-term performance of concrete structures under various environmental conditions.



(a)Effect of temperature on model prediction (b)Effect of relative humidity on model prediction

Fig. 19. Sensitivity analysis of the GSA model to temperature and relative humidity variations.

5.4. Sensitivity analysis of the model to temperature and humidity variations

To verify the sensitivity of the GSA model to temperature and humidity variations, the recursive multi-step forecasting was used to calculate how predictions vary under different temperatures and relative humidity, as shown in Fig. 19. In the temperature impact analysis, concrete strength, loading age, and relative humidity were set at 15.5 MPa, 28 days, and 99 %, respectively. In the humidity impact analysis, concrete strength and loading age were set at 45.2 MPa and 8 days, with a constant temperature of 23°C.

Fig. 19 (a) shows that the GSA model's predicted creep coefficient increases significantly with temperature. When the temperature rises from 20°C to 60°C, the 120-day prediction increases by 104 %. Fig. 19 (b) indicates that the predicted shrinkage strain decreases with increasing relative humidity, with a smaller variation compared to creep. When relative humidity increases from 20 % to 75 %, the 360-day prediction decreases by 10.6 %.

It should be noted that the input parameters of temperature and relative humidity for the GSA model are average constant values, as the database was collected under constant indoor conditions. The model's adaptability to varying outdoor temperature and humidity depends on the dynamic adjustment of historical parameters. In simple terms, temperature and humidity affect historical values, which in turn influence current predictions.

6. Conclusion

Based on the NU database, this study constructed a deep network model combining GRU and a self-attention mechanism (GSA model). The performance of the GSA model was validated, and its applicability to experimental results was compared with traditional shrinkage and creep models. Using short-term experimental data and the GSA model, a recursive multi-step method was employed to extrapolate the ten-year shrinkage and creep development trends. The following conclusions can be drawn from the research results:

1. The training results of the GSA model indicate good performance and considerable generalization capability. Compared to the LSTM model, the GSA model converges faster, saving training time. In the validation set, the distribution of residuals and evaluation metrics show that the GSA model outperforms the LSTM model in predictive accuracy.

2. The creep compliance and shrinkage strain of outdoor specimens exhibit significant fluctuations due to the influence of environmental temperature and humidity, with shrinkage strain showing considerable dispersion. Only the GSA model demonstrates a similar fluctuation trend to the experimental curves of creep and shrinkage. This indicates that the GSA model can capture real-time changes by utilizing historical shrinkage and creep data, thus accounting for the effects of fluctuating outdoor temperature and humidity.

3. In the comparison of experimental values and model predictions, the GSA model had the smallest coefficient of variation (V_{CEB}) for both creep and shrinkage tests, with values of 0.06 and 0.61, respectively, indicating the best predictive performance for on-site shrinkage and creep test results. Among traditional creep models, the AASHTO and fib2010 models showed better predictive performance. For shrinkage models, the AASHTO and GL2000 models performed better.

4. Based on six months of shrinkage and creep test data, the GSA model and the recursive multi-step method were used to predict the development trends over the next ten years. After ten years, the GSA predicted value for COC3 was 58.7×10^{-6}/MPa, a 12.8 % increase from 52.0×10^{-6}/MPa at six months. For COC7, the GSA predicted value reached 83.0×10^{-6}/MPa, a 13.6 % increase from 72.9×10^{-6}/MPa at six months. These results indicate that the GSA model has better adaptability and generalization capabilities for the random development of creep caused by variability in concrete parameters.

5. The AASHTO model underestimated the SLC3 test values, while the GL2000 model overestimated the SLC7 test values. Only the GSA model could accurately predict both SLC3 and SLC7 test values, further validating the superior accuracy of the GSA model. The recursive multi-step predictions of the GSA model fell between the AASHTO and GL2000 models and aligned more closely with traditional models that better matched short-term experimental values, indicating that the long-term predictions of the GSA model are reasonable.

CRediT authorship contribution statement

Zihao Fan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yulin Zhan:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Tianyu Xiang:** Writing – review & editing, Methodology, Investigation. **Yuanyuan Huang:** Investigation. **Binghui Li:** Investigation. **Hui Xiang:** Investigation. **Yingxiong Li:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

We declare that we have no financial or personal relationships with other people or organizations that could inappropriately influence our work. There is no professional or other personal interest of any kind in any product, service or company that could be perceived as influencing the content or review of the manuscript entitled.

Data availability

Data will be made available on request.

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